Trend research — 2026-04-19 · incr 24

Only 8% of structure flips are intended spike→channel — 30% cross the bull/bear line, but aggregate damping eats them (r=0.14)

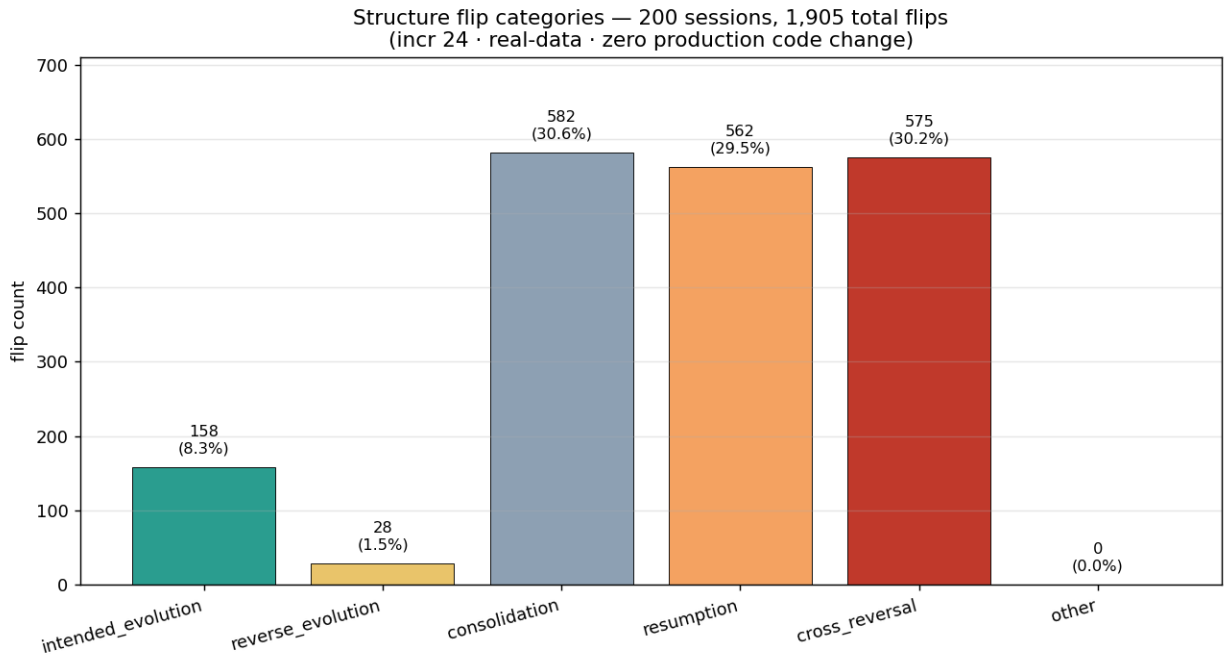
TL;DR

Incr 23 speculated the 1 905 structure flips are mostly the intended Brooks spike → channel progression. Re-reading the persisted 9 425-row trajectory and classifying every flip disproves that: only **8.3 %** are **intended_evolution**. **30.2 %** are **cross_reversal** — the structure label literally crossed the bull/bear boundary. Yet cross_reversal correlates with direction flips at only **r = 0.144**: aggregate direction damping does its job, absorbing ~5 of every 6 structure-level cross-reversals without a direction flip. No production code change. Front-end rule: latch on direction, treat structure as an instantaneous descriptor.

Headline numbers

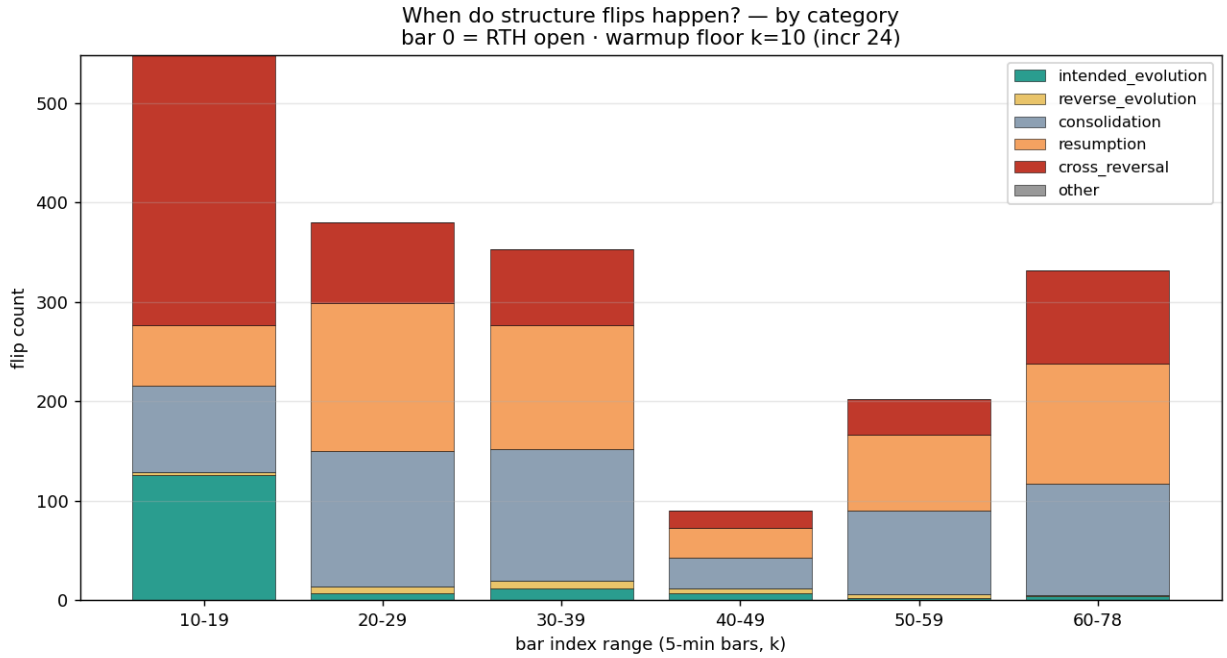
- **1 905 structure flips** across **200 sessions** (9.53 / session, matches incr 23).
- **intended_evolution** (bull_spike → bull_channel, bear_spike → bear_channel): **158 (8.3 %)**. The well-behaved category; **80 %** of them happen in bars 10-19.
- **cross_reversal** (bull_* → bear_* or vice versa): **575 (30.2 %)**. Median **3 / session**, p95 **7 / session**. Bimodal timing: 271 in bars 10-19, then a second bump of 94 in bars 60-78 (end-of-day chop).
- **consolidation** (→ trading_range): **582 (30.6 %)**. **resumption** (trading_range →): **562 (29.5 %)**. Sessions enter and leave ranges at roughly equal rates.
- **reverse_evolution** (channel → spike, same direction): **28 (1.5 %)**. Rare.
- **cross_reversal** ↔ **direction flips: Pearson r = 0.144** (n = 200). Aggregate damping absorbs most structure-level cross-reversals.

Headline figure — flip-category breakdown



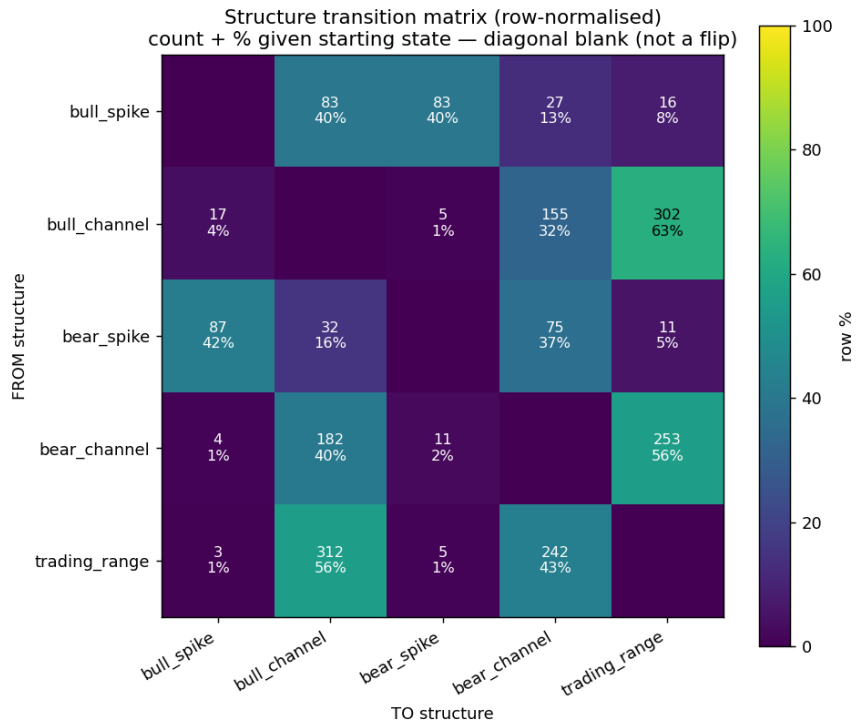
Count and percentage of structure flips in each of 5 mutually-exclusive categories. intended_evolution (teal) is only 8 %. cross_reversal (red), consolidation (grey), and resumption (orange) each take ~30 %. reverse_evolution (yellow) is 1.5 %. The benign-evolution story from incr 23 was true as a distribution snapshot (94 % in channels by bar 78) but false as a description of the path.

Timing — when each category fires



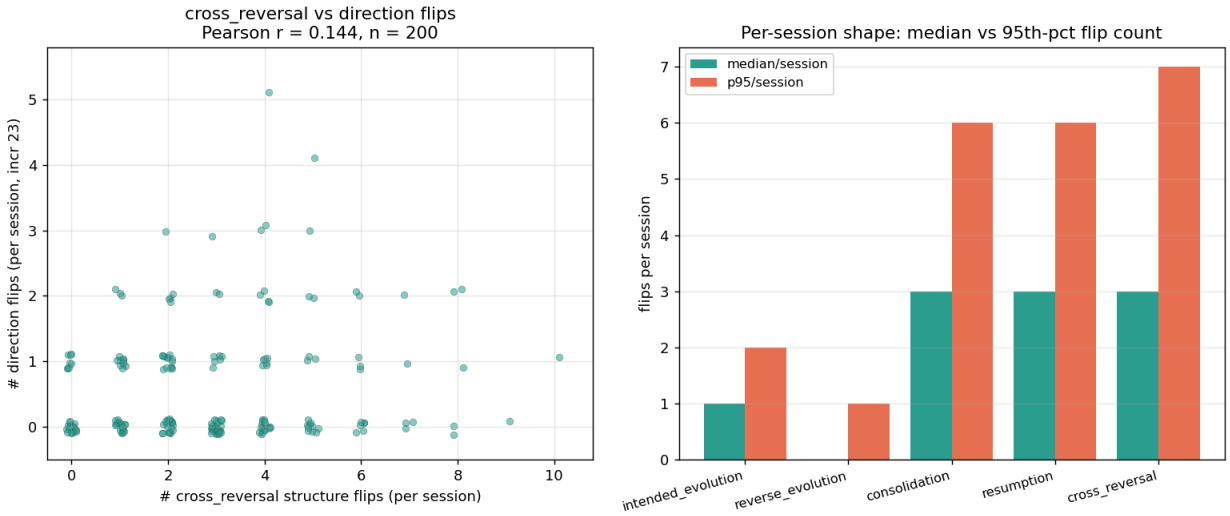
Stacked bars, flips per 10-bar bucket. intended_evolution is front-loaded (bars 10-19). cross_reversal, consolidation, and resumption are heavy at the open AND again at end-of-day — the middle of the session is quieter. This bimodal pattern reinforces incr 23's rule: the safe real-time read is bar 20 with |strength| ≥ 0.15, which dodges most of the 60-78 end-of-day chop.

Transition matrix — where each structure goes next



Row-normalised heatmap: given starting structure X, what fraction of its exits go to each other structure? Trading_range is the single most-visited next state from every channel/spike label (left column of high-intensity cells). Diagonals left blank (same label isn't a flip).

Does structure cross_reversal track direction flips?



Left: per-session scatter of cross_reversal structure flips vs direction flips. Pearson r = 0.144. Most sessions sit on the y = 0 line — meaning lots of structure cross-reversals with zero direction flips. Right: median vs p95 per-session flip counts by category — shows the long tail of consolidation / resumption / cross_reversal, which can spike to 6-7 flips / session on messy days.

Mistakes avoided / corrections

- **Corrected incr 23's prose.** The claim 'structure flips are mostly intended Brooks spike → channel evolution' was based on the end-of-day distribution snapshot, not the flip-count breakdown. Only 8 % of flips actually fit that description.
- **Didn't re-run the progressive-k sweep.** The 9 425-row trajectory CSV from incr 23 is the durable artifact — all flip-type questions answered off it. Zero new Databento calls.
- **Didn't claim cross_reversal = direction flip.** r = 0.144 with direction flips. Structure can cross the bull/bear line while aggregate direction stays put. That's the aggregator working, not a bug.
- **Didn't propose any production change.** The insight only sharpens front-end consumer rules — latch on direction, not structure.
- **Safety-net category `other` stayed at 0.** The 5-label structure alphabet completely partitions; no unclassified transitions.

What's next — requires Will's nod

- **No new recommendation from this pass.** The incr-23 front-end rule ('|strength| ≥ 0.15 at bar 20 → high-confidence direction') is reinforced: gate on direction, not structure.
- **Still pending:** ALWAYS_IN_WINDOW 5 → 10 (incr 20), MAJORITY_TREND_BAR_FLOOR 0.40 → 0.25 (incr 18), OPEN_EXTREME_THRESHOLD 0.15 → 0.25 (incr 21), htf_alignment silent-zero docstring (incr 19), trend_state payload emission.
- **Open follow-up** — Pattern Lab WR-driven weighting decision still waiting on Pattern Lab data to flow.

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification